Robotic Arm Project

Open Meeting – 10/16/25

It’s time to announce teams!

Electrical (Hardware):

- [Project leads will lead this team]
- Juzer Abid,
- Phuong Nguyen,
- Jacob Presas,
- Vanina Joyce Nguemsop Feutie
- Josh Carreon

Electrical (Firmware):

- Jacob Hernandez [Team Lead]
- Roberto Revelo
- Carlos Hernandez
- Joshua Kurian
- Nisa Rodriguez
- Jonathan Pham
- Leziga Beage
- Furqan Muhammed Ahcom

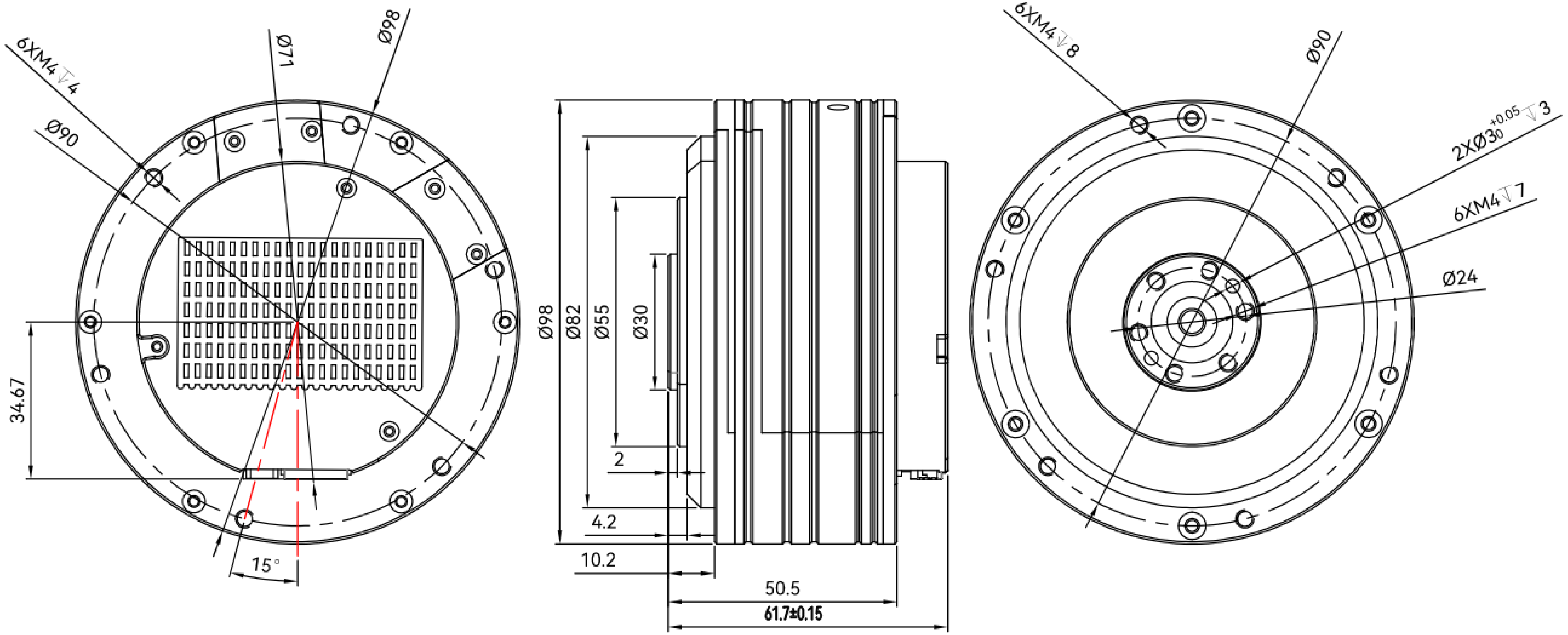
Mechanical:

- Yvette Ayodoro [Team Lead]
- Isela Garcia
- Noah Laclair
- Patrik Wagner
- Christopher Alvarado
- Sarah Lambert
- Victoria DeHoyos

Congrats to everyone selected!

If you have any questions or concerns about your selection, let us know after the meeting or over Discord.

Mechanical Design Task: Model our motor using CAD



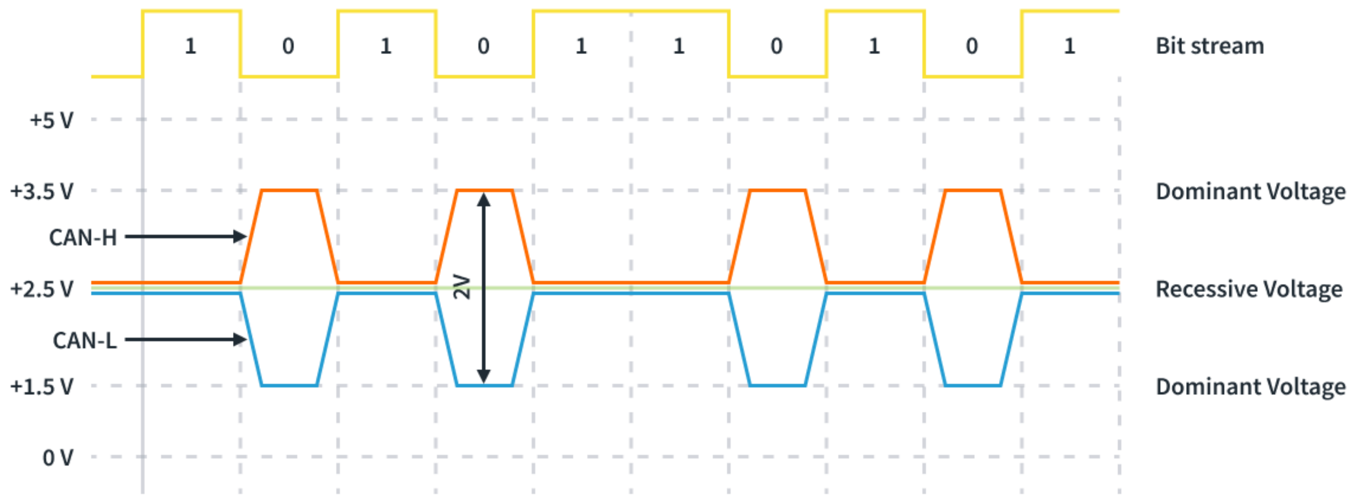
Mechanical Design Task: Model our motor using CAD

Seek to accurately model the motor based on the dimensions provided in the engineering drawing.

We only expect a rough model. The fine details are missing the needed dimensioning to attempt to model.

Advanced Challenge: Allow the motor to spin (requires modeling two parts)

CAN: Controller Area Network



CAN High-Speed (ISO 11898-2) Waveform

CAN uses a differential signal model rather than a simple Hi/Lo signal with a 3.5V CAN-H wire and a 1.5V CAN-L wire.

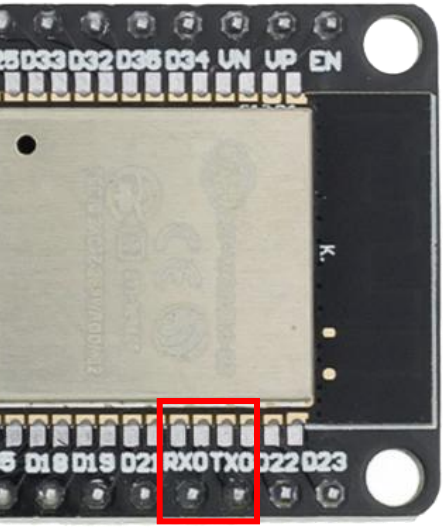
This reduces noise in the signal, resulting in a stable and reliable system.

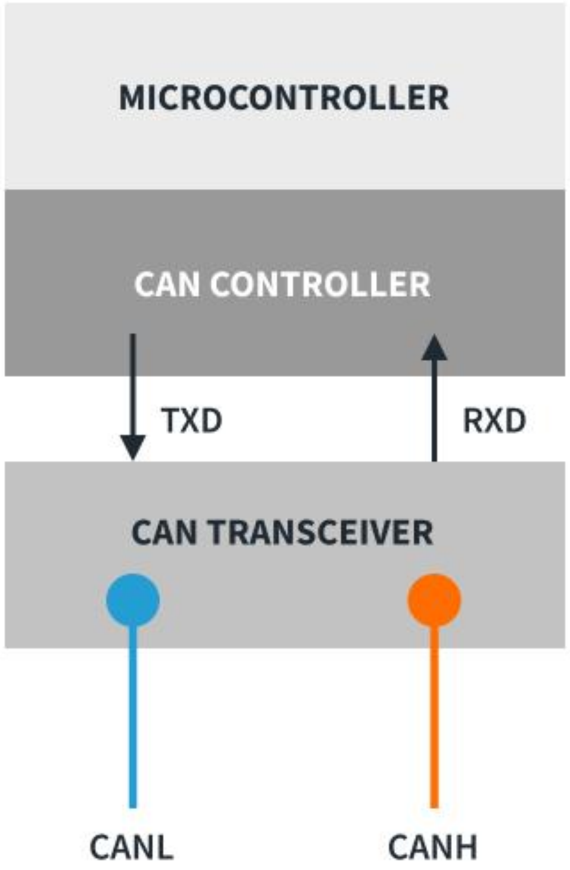
$$V_{ref} = V_{High} - V_{Low}$$

A "0" in the bitstream results from $V_{ref} \approx 2v$

A "1" in the bitstream results from $V_{ref} \approx 0v$

CAN: Controller Area Network





CAN communicates to an MCU/ECU through the TX and RX pins (plus a common GND)

Any device on a CAN network can both transmit and receive data to/from any other device on the network.

CAN uses a packet communication structure, with each device having a unique address.

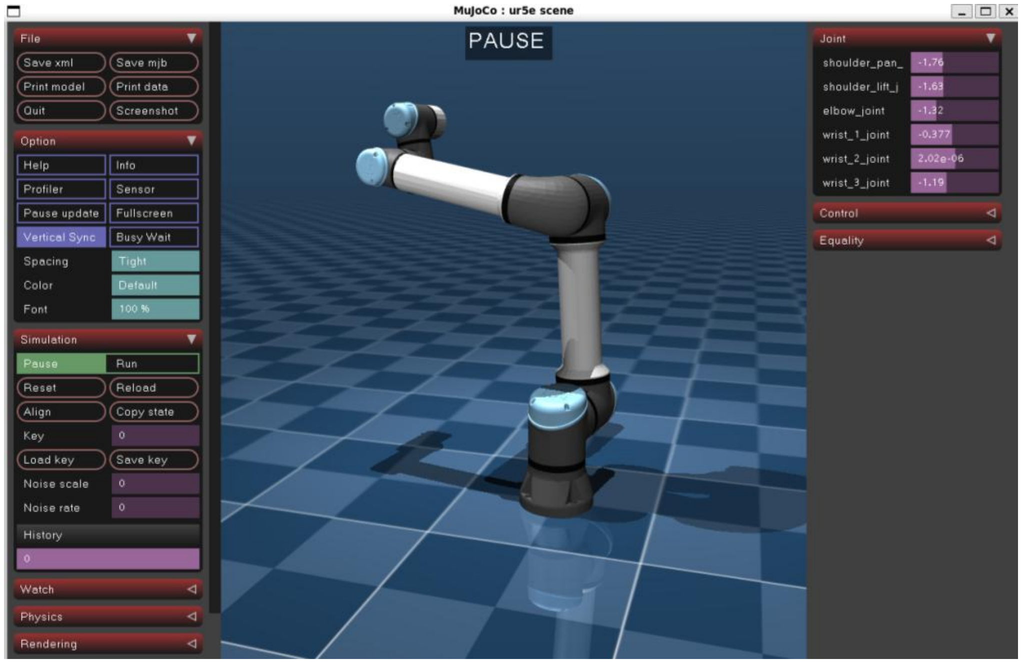
Lost or corrupt packets will be re-transmitted as part of the standard's built-in error correction.

We'll continue reviewing some applications specific to the arm project on the whiteboard

Simulation Team

IsaacSim

- Insert UR5e robot .URDF file into Isaac Sim



MuJoCo

- GitHub tutorial to setup MuJoCo
- Clone model suite mujoco_menagerie
- Run the interactive viewer on:
  mujoco_menagerie/universal_robots/ur5e/scene.xml
- Follow the official YouTube tutorial